COURSE SYLLABUS

BOROUGH OF MANHATTAN COMMUNITY COLLEGE
The City University of New York
Department of Computer Information Systems

Title of Course: Data Structures **Class Hours per Week: 3**

CSC 331 130L Lecture Tue 1:00-1:50 PM FH 608 **Lab Hours per Week: 2**

130B Lab Tue/Thu 2:00-3:40 PM FH 509 **Credits: 3**

Fall 2026

Name:	**Jose Ramon Santos**
Office:	**FH 930 O**
Email:	**josantos@bmcc.cuny.edu**
Phone:	**212-220-8386**
Office Hours:	**Tuesdays 4:00-7:00 PM**

Course Description: This course will introduce students to linear and non-linear data structures, their use and implementation, algorithms, and software engineering techniques. Topics will include: stacks, queues, linked lists, hash tables, trees, graphs, searching and sorting techniques. Asymptotic analysis of algorithms and data structures will also be discussed.

Basic skills:

ENG 095, ESL 095, ACR 095.

Prerequisites:

CSC 211 (Advanced Programming Techniques) and CSC 231 (Discrete Structures) or Departmental approval.

Student Learning Outcomes:

Course Student Learning Outcomes (Students will be able to…)	Measurements (means of assessment for student learning outcomes listed in first column)
1. Demonstrate knowledge of various data structures and implement these structures in a high level language.	**1.** Short essay questions in exams, programming questions in exams and programming projects.
2. Use recursion for the implementation of algorithms.	**2.** Programming projects and exam questions.
3. Analyze and compare the performance of various sorting algorithms in terms of "Big-O" complexity and space requirements and be able to implement the algorithms.	**3.** Exam questions and programming projects.
5. Analyze a problem and determine the appropriate data structure for the solution of the problem.	**5.** Exams questions and programming projects.

General Education Outcomes and Assessment:

	General Education Learning Outcomes	Measurements (means of assessment for general education goals listed in first column)
X	**Communication Skills-** Students will be able to write, read, listen and speak critically and effectively.	Provide documentation for all programming projects and short essay questions.
X	**Quantitative Reasoning-** Students will be able to use quantitative skills and the concepts and methods of mathematics to solve problems.	Use formulas and concepts of mathematics to solve problems in programming assignments and exams.
	Scientific Reasoning- Students will be able to apply the concepts and methods of the natural sciences.	
	Social and Behavioral Sciences- Students will be able to apply the concepts and methods of the social sciences.	
	Arts & Humanities- Students will be able to develop knowledge and understanding of the arts and literature through critiques of works of art, music, theatre or literature.	
X	**Information & Technology Literacy-** Students will be able to collect, evaluate and interpret information and effectively use information technologies.	Use a high-level computer programming language to create application software.
	Values- Students will be able to make informed choices based on an understanding of personal values, human diversity, multicultural awareness and social responsibility.	

Required Text & Readings

Textbook **Data Structures using C++, Second Edition**
Author: **D.S. Malik**
Publisher: **Course Technology, Cengage Learning**
ISBN-13: **978-0-324-78201-1**
ISBN-10: **0-324-78201-1**

Supplemental Textbook

C++ Programming: Program Design Including Data Structures, 8th Edition, D.S. Malik, Cengage Learning
ISBN-13: 9781337117562 ©2018

Other Resources
Flash drives are recommended**.**

Evaluation & Requirements of Students:

Midterm	30%
Final Exam	30%
Participation*	10%
Homework and Projects	30%
Total	100%

*Participation includes engaging in class activities and lab exercises and attending SI Leader sessions.

Outline of Topics

Week No.	Topics
Week 1	Overview/Review
Week 2	Array-Based Lists
Week 3	Linked lists I
Week 4	Linked Lists II
Week 5	Recursion
Week 6	Stacks and Queues
Week 7	Queues and MIDTERM
Week 8	Searching and Hashing
Week 9	Sorting Algorithms
Week 10	Sorting Algorithms
Week 11	Binary Trees
Week 12	Binary Search Trees and AVL Trees
Week 13	Graphs I
Week 14	Graphs II and Review
Week 15	Final Exam

Class Participation
Participation in the academic activity of each course is a significant component of the learning process and plays a major role in determining overall student academic achievement. Academic activities may include, but are not limited to, attending class, submitting assignments, engaging in in-class or online activities, taking exams, and/or participating in group work. Each instructor has the right to establish their own class participation policy, and it is each student's responsibility to be familiar with and follow the participation policies for each course.

BMCC is committed to the health and well-being of all students. It is common for everyone to seek assistance at some point in their life, and there are free and confidential services on campus that can help.

Single Stop www.bmcc.cuny.edu/singlestop, room S230, 212-220-8195, singlestop@bmcc.cuny.edu. If you are having problems with food or housing insecurity, finances, health insurance or anything else that might get in the way of your studies at BMCC, come by the Single Stop Office for advice and assistance. Assistance is also available through the Office of Student Affairs, S350, 212-220-8130, studentaffairs@bmcc.cuny.edu.
Counseling Center www.bmcc.cuny.edu/counseling, room S343, 212-220-8140, counselingcenter@bmcc.cuny.edu. Counselors assist students in addressing psychological and adjustment issues (i.e., depression, anxiety, and relationships) and can help with stress, time management and more. Counselors are available for walk-in visits.

Office of Compliance and Diversity https://www.bmcc.cuny.edu/about-bmcc/compliance-diversity, room S701, 212-220-1236. BMCC is committed to promoting a diverse and inclusive learning environment free of unlawful discrimination/harassment, including sexual harassment, where all students are treated fairly. For information about BMCC's policies and resources, or to request additional assistance in this area, please visit or call the office, or email olevy@bmcc.cuny.edu, or twade@bmcc.cuny.edu. If you need immediate assistance, please contact BMCC Public safety at 212-220-8080.
Office of Accessibility www.bmcc.cuny.edu/accessibility, room N360 (accessible entrance: 77 Harrison Street), 212-220-8180, accessibility@bmcc.cuny.edu. This office collaborates with students who have documented disabilities, to coordinate support services, reasonable accommodations, and programs that enable equal access to education and college life. To request an accommodation due to a documented disability, please visit, call the office or email.
BMCC Policy on Plagiarism and Academic Integrity Statement
Plagiarism is the presentation of someone else's ideas, words or artistic, scientific, or technical work as one's own creation. Using the idea or work of another is permissible only when the original author is identified. Paraphrasing and summarizing, as well as direct quotations, require citations to the original source. Plagiarism may be intentional or unintentional. Lack of dishonest intent does not necessarily absolve a student of responsibility for plagiarism. Students who are unsure how and when to provide documentation are advised to consult with their instructors. The library has guides designed to help students to appropriately identify a cited work. The full policy can be found on BMCC's Web site, www.bmcc.cuny.edu. For further information on integrity and behavior, please consult the college bulletin (also available online).